An Introduction to an Open Source Verilog Simulator

Verilog, a hardware description language, is used to model electronic systems, and to design and verify digital circuitry at the register-transfer level of abstraction. It can also be used to verify analogue circuits and mixed signal circuits.

Computers now play a major role in every design field. The complexity of structures is so high that we can no longer think about doing things manually. This is the trend not only in electronics but in all spheres of engineering. Gone are the days when designers could verify their circuits on a breadboard. Today's integrated circuits are so complex that they contain thousands of transistors on a single chip.

In the 1980s, designers felt the need for a language that could model digital circuits so that the circuit could be simulated and analysed. That's how hardware description languages (HDL) came into being.

In this article, I will introduce you to an open source Verilog simulation and synthesiser tool. It is assumed that the readers have some basic knowledge in Verilog programming.

Design flow

HDLs alone were not sufficient to reduce the workload on VLSI circuit designers. Soon, logic synthesis tools were developed, which could translate an HDL-based design into a schematic circuit.

Figure 1 gives a brief idea about the process flow in VLSI IC design. First the requirements of the circuit to be designed are studied, and the designers prepare a behavioural model that will describe the functional requirements of the circuit. In the HDL, both behavioural and architectural details are taken care of. Then the logic synthesis tools convert the HDL to a gate level netlist.

Need for HDL

Hardware description languages differ from software programming languages because they include ways of describing the propagation of time and signal dependencies. With HDL, the designs are made at a very abstract level, which are independent of the technology that will be used for IC fabrication. This helps in the functional verification of the design and also in optimising it.

Verilog

Verilog was created by Phil Moorby and Prabhu Goel in 1984. It was invented as a simulation language and support for synthesis was only added afterwards. An IEEE working group was established in 1993 under the Design Automation Sub-Committee to produce the IEEE Verilog standard 1364. Verilog became IEEE Standard 1364 in 1995. The IEEE working group released a revised standard in March 2002, and it was known as IEEE 1364-2001. Significant publication errors marred this release, and a revised version was released in 2003, which was known as IEEE 1364-2001 Revision C.

VHDL

VHDL is an acronym for Very High Speed Integrated Circuit Hardware Description Language, which is a programming language used to describe a logic circuit by function, data flow behaviour or structure. The development of VHDL was initiated in 1981 by the United States Department of Defence.

Icarus Verilog compiler and GTKWave

Icarus Verilog or iverilog is an implementation of the Verilog hardware description language. It supports the 1995, 2001 and 2005 versions of the standard, portions of SystemVerilog and some extensions.

Icarus Verilog is available for Linux, Windows and Mac OS X. It is released under the GNU General Public License.